Contents

MIOP terms

methodology_category: omics analysis project: “NOAA Atlantic Oceanographic and Meteorological Laboratory Omics Program; https://github.com/aomlomics/protocols; https://zenodo.org/communities/aomlomics” purpose: PCR [OBI:0000415], Library Preparation [OBI:0000711] analyses: PCR [OBI:0000415], Library Preparation [OBI:0000711] geographic_location: Atlantic Ocean [GAZ:00000344], Gulf of Mexico [GAZ:00002853] broad_scale_environmental_context: marine biome [ENVO:00000447], marine photic zone [ENVO:00000209] local_environmental_context: marine biome [ENVO:00000447], marine photic zone [ENVO:00000209] environmental_medium: sea water [ENVO:00002149], polymerase chain reaction [OBI:0000415] target: whole genome sequencing [OBI_0002117] creator: Luke Thompson materials_required: vortexer [OBI:0400118], PCR instrument [OBI:0000989], Centrifuge [OBI:0400106] skills_required: sterile technique, pipetting skills, standard molecular technique time_required: 200 personnel_required: 1 language: en issued: 2026-05-08 audience: scientists publisher: NOAA Atlantic Oceanographic and Meteorological Laboratory hasVersion: 1 license: CC0 1.0 Universal maturity level: mature

FAIRe terms

barcoding_pcr_appr: two-step pcr library_id: not applicable platform: AVITI instrument: seq_kit: lib_layout: paired end sequencing_location: adapter_forward: adapter_reverse: lib_screen: mid_forward: mid_reverse: filename: filename2: seq_method_additional: not applicable

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NOAA/AOML Amplicon Library Prep Protocol

1 PROTOCOL INFORMATION

1.1 Minimum Information about an Omics Protocol (MIOP)

- MIOP terms are listed in the YAML frontmatter of this page.
- See https://github.com/BeBOP-OBON/miop/blob/main/model/schema/terms.yaml for list and definitions.

1.2 Making eDNA FAIR (FAIRe)

- FAIRe terms are listed in the YAML frontmatter of this page.
- See https://fair-edna.github.io/download.html for the FAIRe checklist and more information.
- See https://fair-edna.github.io/guidelines.html#missing-values for guidelines on missing values that can be used for missing FAIRe or MIOP terms.

1.3 Authors

PREPARED BY	AFFILIATION	ORCID	DATE
Luke Thompson	NOAA/AOML, MSU/NGI	https://orcid.org/0000-0002-3911-1280	2026-05-11
Sammy Harding	NOAA/AOML, MSU/NGI	https://orcid.org/0009-0008-8885-6140	2026-05-11

1.4 Related Protocols

- This section contains protocols that should be known to users of this protocol.
- Internal Protocols: Derivative or altered protocols, or other protocols in this workflow.
- External Protocols: Protcols from manufacturers or other groups.
- Include the link to each protocol.
- Include the version number (internal) or access date (external) of the protocol when it was accessed.

1.4.1 Internal Protocols

PROTOCOL NAME	LINK	VERSION	RELEASE DATE
AOML 'Omics Protocols	https://github.com/aomlomics/protocols	not applicable	ongoing
NOAA 'Omics Metabarcoding Assays	https://github.com/NOAA-Omics/noaa-omics-metabarcoding-assays	not applicable	ongoing

1.4.2 External Protocols

PROTOCOL NAME	LINK	ISSUER / AUTHOR	ACCESS DATE
Just-a-Plate 96 PCR Purification and Normalization Kit Manual	http://www.wuhancharmbiotech.com/product/upload/20150630/1435646883.pdf	Charm Biotech	2026-05-08
Quant-iT™ dsDNA High-Sensitivity Assay Kit Manual	https://documents.thermofisher.com/TFS-Assets/LSG/manuals/Quant_iT_dsDNA_HS_Assay_UG.pdf	Invitrogen	2026-05-08
D1000 ScreenTape Assay for TapeStation Systems Manual	https://www.agilent.com/cs/library/usermanuals/public/D1000_QuickGuide.pdf?srsltid=Afm	Agilent	2026-05-08

1.5 Protocol Revision Record

VERSION	RELEASE DATE	DESCRIPTION OF REVISIONS
1.0.0	2026-05-11	Initial release

1.6 Acronyms and Abbreviations

ACRONYM / ABBREVIATION	DEFINITION
NOAA	National Oceanographic and Atmospheric Administration
AOML	Atlantic Oceanographic and Meteorological Laboratory
MSU	Mississippi State University
NGI	Northern Gulf Institute
PCR	Polymerase chain reaction
eDNA	environmental DNA
NTC	No template control
EtOH	Ethanol

2 BACKGROUND

2.1 Summary

This protocol was developed and implemented by NOAA AOML's Molecular and Computational Biodiversity group to generate Illumina-compatible environmental DNA (eDNA) amplicon libraries using the Fluidigm CS1/CS2 indexing system, enabling conversion of primary PCR amplicons into uniquely dual-indexed libraries suitable for AVITI sequencing. The workflow consists of a secondary indexing PCR, bead-free normalization and purification, pooled library quality control and final validation of library concentration and fragment size before sequencing submission.

2.2 Method description and rationale

This protocol uses a secondary PCR to attach Illumina-compatible adapters and unique dual indexes to CS1/CS2-tagged eDNA amplicons, enabling multiplexed sequencing on the AVITI platform. Following indexing, libraries are purified and normalized using the Just-a-Plate™ system to remove residual primers and equalize sample concentrations prior to pooling. The pooled library is then quantified fluorometrically and assessed on the TapeStation to confirm appropriate concentration and fragment size distribution while identifying potential primer-dimer contamination. The two-step PCR design improves amplification consistency and reduces costs associated with fully indexed locus-specific primers, while unique dual indexing minimizes sample misassignment and index hopping. Overall, the workflow is optimized for high-throughput, reproducible preparation of eDNA amplicon libraries suitable for downstream biodiversity and metabarcoding analyses.

2.3 Spatial coverage and environment(s) of relevance

This protocol is intended for environmental DNA (eDNA) metabarcoding applications in marine environments to support high-throughput biodiversity monitoring and ecological assessment.

3 PERSONNEL REQUIRED

One person with molecular biology experience.

3.1 Safety

There are no hazardous chemicals or materials involved in this protocol. Standard lab safety techniques should still be used such as wearing PPE to avoid skin or eye contact.

3.2 Training requirements

Basic molecular biology training is sufficient for this protocol including sterile technique, pipetting small volumes and programming/running PCR thermal cyclers.

3.3 Time needed to execute the procedure

This protocol takes about 3.5 hours to complete.

4 EQUIPMENT

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
Durable equipment				
Thermal cycler	Mastercycler Nexus Thermal Cycler	Eppendorf	1	Can be substituted with generic
Vortex	Vortex Genie	Scientific Industries	1	Can be substituted with generic
PCR Plate Centrifuge	Microplate Centrifuge	Generic Brand	1	Can be substituted with generic
Agilent 4200 TapeStation	Agilent 4200 TapeStation	Agilent	1	
0.5-10 ul Pipette	Eppendorf Research Plus Adjustable-Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette
100-1000 ul Pipette	Eppendorf Research Plus Adjustable-Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
10-100 ul Pipette	Eppendorf Research Plus Adjustable-Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette
10-100 ul 8-Channel Pipette	Eppendorf Research Plus 8 Channel Pipette	Eppendorf	1	Can be substituted with any accurate pipette
0.5-10 uL 8-Channel Pipette	Eppendorf Research Plus 8 Channel Pipette	Eppendorf	1	Can be substituted with any accurate pipette
Serological Pipette	Eppendorf Easypet 3 Pipette	Eppendorf	1	Can be substituted with any accurate pipette
Consumable equipment				
DreamTaq Master Mix	DreamTaq PCR Master Mix (2X)	ThermoFisher	1	
Just-a-Plate Kit	Just-a-Plate ™ 96 PCR Normalization and Purification Kit, 96 X 10 (Preps)	CharmBiotech	1	
Barcode Library	Access Array™ Barcode Library for Illumina® Sequencers— 384, Single Direction	Standard BioTools	1	
Quant-iT kit	Invitrogen Quant-iT 1X dsDNA Assay Kits, high sensitivity (HS) and broad range (BR)	Invitrogen	1	
TapeStation Assay	TapeStation D1000 DNA ScreenTape & Reagents	Agilent	1	
96-well PCR plate	Armadillo PCR Plate, 96-well, clear, clear wells	ThermoFisher	1	
Aluminum Foil Sealing Tape	AlumaSeal II Sealing Foils for PCR and Cold Storage	VWR	7	Can be substituted with generic

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
8-Strip Tubes	0.2 mL 8-Strip PCR Tubes	Generic Brand	5	tubes without caps attached are better
Microcentrifuge Tubes	2.0 mL Microcentrifuge Tube	Generic Brand	8	Can be substituted with generic
1000µL Filter Tips	OT-2 Filter Tips, 1000µL	Opentrons	1	(box) Can be substituted with generic
200µL Filter Tips	OT-2 Filter Tips, 200µL	Opentrons	7	(boxes) Can be substituted with generic
10 ul Filter tips	TipOne Pipette Tips, 10 uL	TipOne	9	(boxes) Can be substituted with generic
Gloves	Nitrile Gloves, Exam Grade, Powder-free	ULINE	1	(box) Can be substituted with generic
Kim Wipes	KimWipe Delicate Task Wipers	KimTech	1	(box) Can be substituted with generic
Chemicals				
80% Ethanol	Molecular biology grade ethanol			
Nuclease-free water	Nuclease-Free Water (not DEPC-Treated)	ThermoFisher Scientific	1	(mL vial)

- Description: E.g., “filter”.
- Product Name and Model: Provide the official name of the product.
- Manufacturer: Provide the name of the manufacturer of the product.
- Quantity: Provide quantities necessary for one application of the standard operating procedure (e.g., number of filters).
- Remark: For example, some of the consumable may need to be sterilized, some commercial solution may need to be diluted or shielded from light during the operating procedure.

5 STANDARD OPERATING PROCEDURE

6 Protocol

6.1 Preparation

1. Enter and save the following protocol on the thermal cycler:

PCR step	Temperature	Duration	Repetition
Initial Denaturation	95°C	3min	1x
Denaturation	95°C	15s	11x
Annealing	60°C	30s	11x
Extension	72°C	90s	11x
Final Extension	72°C	3min	1x

PCR step	Temperature	Duration	Repetition
Hold	4°C	∞	

6.2 Step 1: Secondary (Indexing) PCR

This step attaches the P5/P7 adapters and i5/i7 barcodes to your primary amplicons.

1. Allow reagents to thaw on ice for 30 minutes.
2. Prepare a master mix (excluding index primers and template DNA):

   For one well:

   - 7.5 uL 2X DreamTaq Master Mix
   - 5.0 uL Nuclease-Free Water

   For a 96-well plate:

   - 750 uL 2X DreamTaq Master Mix
   - 500 uL Nuclease-Free Water

3. Pipette 12.5 uL of master mix into each well of a PCR plate.
4. Using a multichannel pipette, carefully pipette 1.5 uL of each barcode primer into respective well of PCR plate and gently pipette to mix.
5. Using a multichannel pipette, carefully pipette 1.0 uL of primary PCR product (or molecular-grade water for NTCs) into respective well of PCR plate so the final reaction volume is 15 uL total.
6. Seal plate with foil and briefly spin down.
7. Place plate on thermal cycler and run the programmed protocol.

6.3 Step 2: Normalization and Purification

This step uses the Just-a-Plate™ system to simultaneously clean and equalize the concentration of your samples.

Binding of PCR Products to Binding Plate

1. Remove PCR2 plate from the thermal cycler and spin down so there is no remaining condensation along the sides of the wells. Allow plate to come to room temp.
2. Using a multichannel, transfer 10-15 µL of PCR2 product to the Binding Plate (BP5).
3. Gently swirl the Binding Buffer (NB7) to mix and add 20 uL of binding buffer to each sample well. Gently pipette to mix 5-6 times (Note: avoid introducing bubbles while pipetting).
4. Seal the Binding Plate (BP5) with a foil and incubate at room temp for 30-60 minutes. (Note: the plate can be incubated overnight at room temperature if it fits your workflow).
5. Remove the plate foil.
6. Remove the liquid from the Binding Plate (BP5).

Washing Plate

7. Prepare working Washing Buffer (WB2) by adding 96 ml of 100% ethanol to bottle and use a sharpie to note that ethanol has been added.
8. Add 50 ul of Wash Buffer (WB2) containing ethanol to each well. Pipette to mix 1-2 times.
9. Incubate the plate for 30 seconds at room temp.
10. Remove the Wash Buffer (WB2) from the Binding Plate (BP5).

11. Repeats step 8, 9 and 10 once more for a total of two washes.
12. After the final wash, blot the plate dry on a piece of clean absorbent paper and air-dry the plate in a lab hood for 5-10 minutes to remove any residual liquid.

Eluting Products

13. Add 20 ul of Elution Buffer (EB1) into each well of the Binding Plate (BP5).
14. Seal the plate with a new plate foil and vortex the plate for 30 seconds. Centrifuge the plate at maximum speed briefly to collect all liquid at the bottom of the wells.
15. The eluted DNA fragments may be used immediately for downstream NGS application. Alternatively, the eluted products may be stored in the sealed plate at 4°C for short-term storage or at –20°C for long-term storage.

6.4 Step 3: Pooling and Quality Control

Pooling

1. Label a 2 mL microcentrifuge tube with the name and info of your final pool.
2. Transfer 5 uL from every well of your normalized plate into tube (the “Final Pool”).

Quantification on Quant-iT

1. Allow Quant-iT reagents to come to room temperature (30 min).
2. Use the first two columns of the Quant-iT plate for replicates of reach of the 8 standards. Pipette 10 uL of each standard into the first two columns.
3. Using a multichannel, pipette 2 uL of each sample into corresponding wells of plate.
4. Using a repeater pipette, carefully dispense 200 uL of working solution into each well of the plate.
5. Turn on varioskan, select correct protocol, place plate onto varioskan and begin run.
    - Target yield: >5 nM (typically ~2–5 ng/µL for most amplicons).

Size Verification on TapeStation

1. Allow HS D1000 tape and reagents to come to room temperature (30 min).
2. Turn on TapeStation, flick tape to remove any air bubbles and place tape into machine.
3. Using an 8-strip tube, pipette the following quantities of reagents:
    - Well A1 (always reserved for ladder): 3 uL buffer + 1 uL ladder DNA
    - Remainder wells: 3 uL buffer + 1 uL sample DNA
4. Briefly spin down 8-strip tube to ensure all volume is in the bottom of the wells.
5. Using plate vortexer, place 8 strip tubes with cap strip into foam and vortex for 1 minute.
6. Remove 8 strip cap and place 8 strip tube onto tapestation.
7. Begin run on TapeStation program and record measurements.

6.5 Basic troubleshooting guide

Not applicable

7 REFERENCES

Not applicable.

8 APPENDIX A: DATASHEETS

Not applicable.